

Grove shield is a modular, simple-to-use system designed to easily connect a BLE development kit to a wide range of “modules” such as sensors, motors, and inputs.

Key Features

- Plug-and-play modular system – easy connection of sensors, actuators, and inputs
- Standardized Grove interface – connectors for I²C, UART, digital, and analog signals
- Compatible with BLE development kits – simplifies prototyping with wireless modules
- No soldering required – rapid prototyping and project development
- Scalable design – supports multiple Grove modules simultaneously



Fig: Grove Shield

Technical Specifications

Parameter	Value
Operating Voltage	3.3 V / 5 V (depends on development kit)
Interfaces Supported	Digital, Analog, I²C, and UART via Grove connectors
Compatibility	Works with Arduino, BLE kits, and Seeed Studio Grove ecosystem
Form Factor	Shield/hat design for easy stacking onto development boards

Applications

- Rapid Prototyping – quickly test sensors, actuators, and wireless modules without soldering
- IoT Development – connect BLE-enabled boards with Grove sensors for smart home or industrial IoT
- Wearables & Health Monitoring – integrate bio-sensors (heart rate, SpO₂, temperature) with BLE kits
- STEM Education & Learning – ideal for teaching electronics and IoT concepts with modular plug-and-play design